

Roger Richard Demello

Nagpur, India

+91 94217 78898 | rogerdemello289@gmail.com | linkedin.com/in/rogerdemello | github.com/rogerdemello | rogerdemello.tech

Summary

AI Engineer specializing in **LLM applications**, **RAG pipelines**, and **Agentic AI systems**. Built and deployed **3+ production-grade AI solutions** serving **200+ users**, reducing latency by **35%**, improving workflow efficiency by **40%**, and accelerating delivery cycles by **50%**. Experienced with **Python**, **FastAPI**, **LangChain**, **vector databases**, and end-to-end AI product development.

Technical Skills

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Prompt Engineering, Agentic AI, Multi-Agent Systems, Embedding Models
Machine Learning & Deep Learning: PyTorch, Scikit-learn, NLP, Semantic Search, Feature Engineering, Computer Vision
Vector Databases & AI Infrastructure: FAISS, Pinecone, ChromaDB, AI System Design, Model Deployment
Backend & DevOps: MLOps, FastAPI, Flask, REST APIs, PostgreSQL, Docker, CI/CD, GitHub Actions, Vercel
Cloud & Tools: AWS, OpenAI, Git, HubSpot API, Hugging Face Transformers, LangChain, Supabase
Programming Languages: C++, Python, Java, C, SQL

Experience

- AI LifeBOT

Jan 2026 - Present

AI Engineer Intern

 - Engineered **3 production LLM applications** and autonomous AI agents (LangChain, GPT-4) serving **200+ users**, increasing task-completion efficiency by **40%** via tool integration, long-term memory, and goal-driven agentic workflows.
 - Architected end-to-end **Generative AI pipelines** (RAG, vector databases, external APIs, real-time streaming), reducing response latency by **35%** and sustaining **99.5% uptime**; designed multi-agent orchestration with approval gates, telemetry pipelines, and audit logging for reproducible agent evaluation.
 - Led prototyping of **5+ AI product features** using Prompt Engineering and Agentic AI design patterns, compressing delivery and validation cycles by **50%** through cross-functional collaboration.
- CFM, RCOEM

May 2025 - Jul 2025

Machine Learning Research Intern

 - Developed a **clinical sleep disorder prediction system** trained on **1,000+ patient health records** with biometric feature engineering (blood pressure, sleep duration, stress indices), achieving **87% prediction accuracy**.
 - Optimized end-to-end **ML data pipelines** (Python, Scikit-learn, Pandas) via cross-validation and rigorous model evaluation, cutting development time by **30%** and boosting validation accuracy by **25%**.

Projects

- Autonomous Executive Email Copilot | Multi-Agent RAG, LangChain, FastAPI, Vector DB | [GitHub](#)

 - Engineered a **multi-agent Agentic AI framework** with a deterministic RL-style simulation environment for benchmarking autonomous email agents across classification, prioritization, and action-execution workflows.
 - Implemented **RAG-powered context-aware prioritization** via vector retrieval and semantic search, achieving an estimated **60% reduction** in inbox triage time through policy-driven decision making.
 - Built **FastAPI telemetry pipelines**, approval gates, and structured audit trails enabling reproducible agent evaluation and real-time AI performance monitoring.
- DealSentry - Intelligent Proposal Guard | Azure OpenAI, RAG, React, TypeScript, PostgreSQL | [GitHub](#)

 - Built an **enterprise AI compliance platform** (Azure OpenAI, RAG) automating proposal validation with AI-driven risk scoring and DOCX/PDF ingestion, reducing manual review effort by **70%**.
 - Designed a **contract lifecycle tracking engine** with rule-based risk scoring, approval workflows, and audit trails supporting full enterprise governance requirements.
 - Integrated **HubSpot and Gmail APIs** to automate end-to-end approval workflows, eliminating manual handoffs and accelerating deal-review cycle time across sales and compliance teams.
- SentinelOps - AI-Powered Incident Detection Platform | FastAPI, Anomaly Detection, Multi-Agent AI | [GitHub](#)

 - Architected an **autonomous incident detection and auto-remediation system** (FastAPI, anomaly detection models) processing **10,000+ synthetic telemetry events** via multi-agent root-cause analysis workflows.
 - Reduced estimated MTTD by **80%** vs. manual monitoring through intelligent anomaly correlation and automated multi-agent impact assessment and remediation recommendations.
 - Integrated **Slack and PagerDuty alerting** with observability pipelines and structured audit logging for proactive infrastructure monitoring and incident correlation.

Education

- Shri Ramdeobaba College of Engineering and Management (RCOEM)

2022 - 2026

B.Tech, Electronics & Communication Engineering

GPA: 8.90/10

Minor: Artificial Intelligence & Machine Learning

GPA: 9.67/10

Certifications & Awards

- AWS Certified Cloud Practitioner** - Amazon Web Services (Oct 2025)
Hackathon Finalist - Paytm x Sarvam x Logitech AI National Hackathon
2nd Place - Bytesage AI National Hackathon